

Spiral-Time as readout and debounced comparator

The shared correction factor $K_{\text{frak}} = 1 - 100\xi$ (74/75) (Doc. 300)

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Doc. 300 Spiral-Time as readout and debounced comparator — and the shared correction factor

$$K_{\text{frak}} = 1 - 100\xi(74/75)$$

Occasion and scope

From the collaboration comes the observation that Spiral-Time is an operator class with at least two roles: a *readout* (what happens in the signal or carrier) and a second part deciding whether this readout may count as a stable transition. This document makes the second role precise in measurement/circuit terms and assigns the correction factor connecting both sides to its *actual* origin in the corpus. Scope: category-B finding (genuine compatibility / candidate overlap), not an HLV-internal derivation, not a physical validation.

The two roles, made precise

The first role, the *readout*, is a measured value: the signal at the +input of a comparison. The second role is not an administrative term ("governance") but precisely a **comparator with hysteresis** — a threshold decision whose output carries one bit. Two forms:

- **amplitude hysteresis**: two level thresholds, a dead band on the value axis stabilising against noise at the switching point.
- **time hysteresis (debounce)**: the output accepts an edge only if it *persists* over a time window. The dead band lies on the time axis.

The second role is the time hysteresis. The finite-proxy window ($M = 12$, $\tau = 6$) is a weighted debouncer: a transition counts only once it survives the window. "Stable transition" is exactly the time hysteresis.

Validity condition: reference from a derivation, not a copy

A comparator carries information only if the reference at the –input comes from a *different* source than the signal. If both inputs carry the same quantity, the output is meaningless. This was the circular core of the U3 boundary gate in the first HLV-side implementation: the reference was a copy of the readout window (`hard = hlv.copy()`), so the correlation is 1.0 by construction. The FFGFT-side test (Doc. 299) replaces it with a *derived* witness (BLP back-flow 5.125, Doc. 283), independent of the object and not reproducible by a hard-reset window. Self-reference is admissible when closed by a derivation; circular when closed by a copy.

Why "the same spiral" is not a postulate

The statement about the correction factor presupposes that the projected HLV object and the FFGFT recursion are the *same* spiral. This identity is supported by several independent contact points:

- 2/9: the same redistribution as the output of the Z_3 -circulant diagonalisation — found, not sought.
- D_6 / φ -**icosahedron**: the same object structure, $\tau = \varphi$, all fifteen axis angles $\arccos(1/\sqrt{5}) = 63.435^\circ$.
- **time**: the spiral-time sector as a bounded section of the rolled-out recursion time (R47).

2/9 and D_6 are geometric, mutually independent, both found; under this multiple anchoring “the same spiral” is the only remaining reading (category B).

The closure comma and K_{frak}

The rolled-out recursion closes only after $N = 1/(100\xi_0) = 75$ equal pieces ($\xi = 4/30000$), not after one turn (Doc. 295/297). The approximation “one turn closes” neglects a small factor: 75 turns each shortened by ε give the exact closure, $75(1-\varepsilon) = 74$, hence

$$\varepsilon = \frac{1}{75} = 100\xi = 0.0133\bar{3} \quad (\text{as an angle } 2\pi/75 = 4.8^\circ \text{ per turn}). \quad (.1)$$

The per-step factor of the recursion is therefore

$$K_{\text{frak}} = 1 - \varepsilon = 1 - 100\xi = \frac{74}{75} = 0.9867, \quad (.2)$$

exactly the form the corpus carries throughout (Doc. 133/231/032/285/018/005). The reciprocal length-ratio form (shortened piece $\rightarrow$ full turn) is $75/74 = 1.0135$.

The actual anchor: not the 75, but Doc. 133

The origin of 100ξ — and thus of 75 and K_{frak} — lies one level deeper and is independent:

- The **scale recursion** $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (Doc. 133/275) has per-step factor $K_{\text{frak}} = 1 - 100\xi$; $75 = 1/(100\xi_0)$ is its winding number.
- The **factor** 100 emerges from the RG running of $D_f = 3 - \xi$ over ~ 17 orders of magnitude (Planck $\rightarrow$ electroweak), effectively $D_f^{\text{eff}} \approx 2.973$ (Doc. 133); geometrically $100 = 4 \cdot 5^2$ (Doc. 042).
- K_{frak} is **not arbitrary** but fixed by requiring two independent derivations of the mass ratio m_e/m_μ to agree (Doc. 133, *Unique Determination*).
- $\xi = 4/30000$ itself is anchored via the Higgs relation, α and Koide.

The closure comma thus *re-expresses* the already-anchored 100ξ ; it does not derive it. The 75 is an output of the same recursion, not an independent anchor.

Consequence for the projection (Marcel)

The statement to Marcel therefore remains valid, only correctly booked: *if* his projected, closed turn object is a section of the same recursion (supported by 2/9, D_6 , time), then the conversion between his projection and the rolled-out exact curve

carries exactly the per-step factor $K_{\text{frak}} = 1 - 100\xi = 74/75$ (reciprocally $75/74$) — a factor anchored *independently* (Doc. 133), not derived from the 75 and not freely chosen. It is a consistency condition, not a derivation from the winding number: if the projection absorbs the comma and the object appears cleanly closed, K_{frak} is the conversion re-entering it at the boundary to the SI level (Doc. 298). The embedding ι (Doc. 299) is this K_{frak} conversion here.

The meshing, not the section

The qualifier “a section of the same recursion” is deliberately indefinite. The 75 pieces are equivalent under the shared discrete symmetry: $75 = 3 \cdot 5^2$ carries exactly the two anchors that also generate the common object — the Z_3 threefoldness (the same as in $2/9 = 2/3^2$) and the icosahedral fivefoldness (the same 5 as in $\arccos(1/\sqrt{5})$ and in $100 = 4 \cdot 5^2$, Doc. 042). The symmetry maps the pieces onto one another; none is distinguished.

Hence the question *which* of the 75 pieces the projected window meets is not open but moot: a distinguished assignment is precisely what the symmetry forbids. What is asserted is only the *existence of the meshing* — that the closed object sits as a piece of this partition at all, whose angular structure D_6 fixes and whose count factors as $3 \cdot 5^2$ — not an index assignment.

For K_{frak} this is inconsequential, and it is exactly why the conversion works without identification: $K_{\text{frak}} = 1 - 100\xi$ is the per-step factor of the partition, the *same* across all 75 steps precisely because they are equivalent. It needs no selected piece to be defined; the mere existence of the meshing suffices for the conversion to the rolled-out curve to be $1 - 100\xi$.

Limit and status

The chain is as strong as its anchoring: $2/9$ and D_6 are booked as shared objects (category B), $K_{\text{frak}} = 1 - 100\xi$ is independently anchored via Doc. 133. Comma check script: `Dok300_Skripte/ffgft_300_kfrak_schliessungskomma_pruefung.py`; time-sector counter-run: `Dok299_Skripte/ffgft_299_iota_gegenlauf_winding_locked.py`.